Requirement Report | #1

Generated 2026-10-30 04:52 UTC

Requirement ID	1
Title	[FYP DEMO] Valid sign-in
Requirement Type	Functional
User Priority	High
Suggested Priority	High
Quality Risk Score	Not assessed
Risk Level	Medium

Analysis provenance

AI analysis (v4)

Ambiguity in credential validation rules and error feedback introduces uncertainty in implementation and testing, which could lead to inconsistent user experience or security gaps if not clarified.

The priority is marked as High based on the input data, but no business rationale was supplied; this needs stakeholder confirmation.

Original Requirement

A registered active user signs in with their valid email and password. Successful sign-in opens the user dashboard. Both fields are required; empty email or password is rejected without opening the dashboard.

Ambiguity Findings

[AI Suggestion] [Needs Confirmation] The requirement does not specify what constitutes a valid email format or password complexity rules.

[AI Suggestion] [Needs Confirmation] The requirement does not specify the exact error message or feedback mechanism when sign-in fails or fields are empty.

Conflicting Statements

None recorded.

Missing Information

[AI Suggestion] [Needs Confirmation] Definition of valid email format and password rules.

[AI Suggestion] [Needs Confirmation] Specific error handling behavior or messaging when empty email or password is provided.

Clarification Questions

[AI Suggestion] [Needs Confirmation] What specific criteria define a valid email and password for a registered active user?

[AI Suggestion] [Needs Confirmation] What exact feedback or error indication must be presented to the user when an empty email or password is submitted?

Testability Findings

[AI Suggestion] [Needs Confirmation] Testing valid sign-in is partially untestable without the specific rules for valid email and password data.

[AI Suggestion] [Needs Confirmation] Testing the rejection of empty fields is untestable without a defined error output or interface response specification.

Assumptions

None recorded.

Improved Requirement

[AI Suggestion] [Needs Confirmation] A registered active user signs in with their valid email and password [TO BE CONFIRMED: specify valid email format and password rules]. Successful sign-in opens the user dashboard. Both fields are required; empty email or password is rejected without opening the dashboard [TO BE CONFIRMED: specify error feedback].

Acceptance Criteria

[AI Suggestion] [Needs Confirmation] AC-001: A registered active user provides a valid email and password, resulting in a successful sign-in and opening of the user dashboard.

[AI Suggestion] [Needs Confirmation] AC-002: An empty email field results in rejection of the sign-in attempt without opening the dashboard.

[AI Suggestion] [Needs Confirmation] AC-003: An empty password field results in rejection of the sign-in attempt without opening the dashboard.

Generated Test Scenarios

[AI Suggestion] SC-52d93183059d4e4ca041dd7143dad3b0-001 | Positive: Verify valid sign-in | Requirement version 2

Linked criteria: AC-001

Preconditions

User is registered and active

User has a valid email and password

Test Steps

Enter valid email

Enter valid password

Request sign-in

Expected Result

Sign-in succeeds and the user dashboard opens.

[AI Suggestion] SC-52d93183059d4e4ca041dd7143dad3b0-002 | Negative: Sign-in rejected with empty email | Requirement version 2

Linked criteria: AC-002

Preconditions

User is on the sign-in interface

Test Steps

Leave email field empty

Enter valid password

Request sign-in

Expected Result

Sign-in is rejected and the user dashboard does not open.

[AI Suggestion] SC-52d93183059d4e4ca041dd7143dad3b0-003 | Negative: Sign-in rejected with empty password | Requirement version 2

Linked criteria: AC-003

Preconditions

User is on the sign-in interface

Test Steps

Enter valid email

Leave password field empty

Request sign-in

Expected Result

Sign-in is rejected and the user dashboard does not open.

Traceability

Requirement	REQ-1
Acceptance Criteria	AC-001, AC-002, AC-003
Test Scenarios	SC-52d93183059d4e4ca041dd7143dad3b0-001, SC-52d93183059d4e4ca041dd7143dad3b0-002, SC-52d93183059d4e4ca041dd7143dad3b0-003
Traceability	See versioned evidence below. Historical cases do not validate the current source; recorded links do not prove complete test coverage.

SQA Review

Review Status	Approved
Reviewer Name	Engineer
Review Timestamp	2026-10-30T04:52:23.674Z

Reviewer Notes

Synthetic local rehearsal review, not supervisor or production approval.

REQ-1 - Versioned traceability and Manual execution

Project current | Current content version 2 | [FYP DEMO] Valid sign-in

SQA review: Approved | Authoring state: Approved | Reviewed version: 2

Recorded criterion links do not prove complete coverage. Only Pass is passed; Not Run and Blocked are not passed. Historical cases/results do not validate a new requirement version.

Requirement version 2 (current)

Recorded 2026-10-30T04:52:21.033Z | Actor 2 | saved version

A registered active user signs in with their valid email and password. Successful sign-in opens the user dashboard. Both fields are required; empty email or password is rejected without opening the dashboard.

AC-001: A registered active user provides a valid email and password, resulting in a successful sign-in and opening of the user dashboard.

AC-002: An empty email field results in rejection of the sign-in attempt without opening the dashboard.

AC-003: An empty password field results in rejection of the sign-in attempt without opening the dashboard.

Requirement version 1 (historical)

Recorded 2026-10-30T04:52:12.053Z | Actor 2 | saved version

A registered active user signs in with their valid email and password. Successful sign-in opens the user dashboard. Both fields are required; empty email or password is rejected without opening the dashboard.

TC-1: Verify valid sign-in

Requirement version 2 | Case revision 4 | Approved | active | Unspecified | Priority High

Criterion links: AC-001

Preconditions: User is registered and active; User has a valid email and password

Test data:

1. Enter valid email

2. Enter valid password

3. Request sign-in

Expected: Sign-in succeeds and the user dashboard opens.

Clarification: None recorded

TC-2: Sign-in rejected with empty email

Requirement version 2 | Case revision 3 | Approved | active | Unspecified | Priority High

Criterion links: AC-002

Preconditions: User is on the sign-in interface

Test data:

1. Leave email field empty

2. Enter valid password

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Clarification: None recorded

TC-3: Sign-in rejected with empty password

Requirement version 2 | Case revision 3 | Approved | active | Unspecified | Priority High

Criterion links: AC-003

Preconditions: User is on the sign-in interface

Test data:

1. Enter valid email

2. Leave password field empty

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Clarification: None recorded

Case snapshot #1 - TC-1 revision 1

Requirement version 2 | Successful sign-in with valid credentials | Criteria: AC-001

Preconditions: User is registered and active; User has a valid email and password | Test data:

1. Enter valid email

2. Enter valid password

3. Request sign-in

Expected: Sign-in succeeds and the user dashboard opens.

Case snapshot #2 - TC-2 revision 1

Requirement version 2 | Sign-in rejected with empty email | Criteria: AC-002

Preconditions: User is on the sign-in interface | Test data:

1. Leave email field empty

2. Enter valid password

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Case snapshot #3 - TC-3 revision 1

Requirement version 2 | Sign-in rejected with empty password | Criteria: AC-003

Preconditions: User is on the sign-in interface | Test data:

1. Enter valid email

2. Leave password field empty

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Case snapshot #4 - TC-1 revision 2

Requirement version 2 | Verify valid sign-in | Criteria: AC-001

Preconditions: User is registered and active; User has a valid email and password | Test data:

1. Enter valid email

2. Enter valid password

3. Request sign-in

Expected: Sign-in succeeds and the user dashboard opens.

Case snapshot #5 - TC-1 revision 3

Requirement version 2 | Verify valid sign-in | Criteria: AC-001

Preconditions: User is registered and active; User has a valid email and password | Test data:

1. Enter valid email

2. Enter valid password

3. Request sign-in

Expected: Sign-in succeeds and the user dashboard opens.

Case snapshot #6 - TC-1 revision 4

Requirement version 2 | Verify valid sign-in | Criteria: AC-001

Preconditions: User is registered and active; User has a valid email and password | Test data:

1. Enter valid email

2. Enter valid password

3. Request sign-in

Expected: Sign-in succeeds and the user dashboard opens.

Case snapshot #7 - TC-2 revision 2

Requirement version 2 | Sign-in rejected with empty email | Criteria: AC-002

Preconditions: User is on the sign-in interface | Test data:

1. Leave email field empty

2. Enter valid password

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Case snapshot #8 - TC-2 revision 3

Requirement version 2 | Sign-in rejected with empty email | Criteria: AC-002

Preconditions: User is on the sign-in interface | Test data:

1. Leave email field empty

2. Enter valid password

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Case snapshot #9 - TC-3 revision 2

Requirement version 2 | Sign-in rejected with empty password | Criteria: AC-003

Preconditions: User is on the sign-in interface | Test data:

1. Enter valid email

2. Leave password field empty

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Case snapshot #10 - TC-3 revision 3

Requirement version 2 | Sign-in rejected with empty password | Criteria: AC-003

Preconditions: User is on the sign-in interface | Test data:

1. Enter valid email

2. Leave password field empty

3. Request sign-in

Expected: Sign-in is rejected and the user dashboard does not open.

Version-bound SQA review history

Review #1 | Requirement version 2 | In Review | Engineer (User 3) | 2026-10-30T04:52:23.625Z | Synthetic local rehearsal review, not supervisor or production approval. | Evidence: {'requirement_revision': 10, 'case_version_ids': [1, 4, 5, 6, 2, 7, 8, 3, 9, 10], 'execution_ids': [], 'bug_ids': []}

Review #2 | Requirement version 2 | Approved | Engineer (User 3) | 2026-10-30T04:52:23.674Z | Synthetic local rehearsal review, not supervisor or production approval. | Evidence: {'requirement_revision': 11, 'case_version_ids': [1, 4, 5, 6, 2, 7, 8, 3, 9, 10], 'execution_ids': [], 'bug_ids': []}